

CHERNIHIV, Ukraine

WMO: 331350

Lat: 51.4439N

Lon: 31.2003E

Elev: 141

StdP: 99.64

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.3	-16.5	-22.3	0.5	-18.5	-19.7	0.7	-15.7	9.4	-0.7	8.5	-1.8	2.6	60	0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	31.0	20.4	29.2	19.8	27.5	19.0	21.7	28.4	20.8	27.1	20.0	25.9	3.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.5	14.5	24.7	18.7	13.7	23.7	17.8	13.0	22.8	63.9	28.4	60.7	27.2	57.9	26.0	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.4	6.6	DB	-22.3	33.6	3.5	2.0	-24.8	35.0	-26.8	36.2	-28.8	37.3	-31.4	38.8	(4)
(5)			WB	-22.5	23.2	3.5	1.2	-25.0	24.0	-27.1	24.7	-29.0	25.4	-31.5	26.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	-4.7	-3.7	1.4	9.2	15.0	18.6	20.6	19.3	14.0	7.6	2.0	-2.8	(6)	
	DBStd	10.10	5.81	5.88	4.64	4.25	4.33	3.51	3.16	3.43	3.88	4.18	4.76	5.70	(7)	
	HDD10.0	1926	456	382	269	64	8	0	0	0	9	99	242	397	(8)	
	HDD18.3	3951	714	616	526	274	119	41	14	28	139	334	490	656	(9)	
	CDD10.0	1232	0	0	1	41	164	257	327	289	128	23	2	0	(10)	
	CDD18.3	215	0	0	0	1	17	48	83	59	8	0	0	0	(11)	
	CDH23.3	1902	0	0	0	9	162	383	706	560	81	1	0	0	(12)	
	CDH26.7	544	0	0	0	1	29	88	221	189	16	0	0	0	(13)	
(14) Wind	WSAvg	3.2	3.5	3.5	3.7	3.3	3.1	3.0	2.8	2.7	2.9	3.0	3.3	3.5	(14)	
(15) Precipitation	PrecAvg	609	38	40	35	42	63	63	73	59	53	49	49	45	(15)	
	PrecMax	798	74	82	82	100	131	133	146	124	117	161	102	115	(16)	
	PrecMin	448	11	17	6	3	26	8	23	4	2	1	3	7	(17)	
	PrecStd	76	16	17	18	24	26	34	39	29	36	35	24	28	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.7	8.4	15.8	24.2	29.4	31.1	33.0	34.2	28.7	21.9	14.0	8.9	(19)	
		MCWB	5.2	5.5	9.0	14.5	18.7	19.9	21.5	21.6	17.6	14.7	11.6	7.8	(20)	
	2%	DB	4.7	5.9	12.8	21.3	27.1	29.2	31.0	31.1	25.7	18.4	11.5	6.8	(21)	
		MCWB	3.4	4.1	7.4	12.9	17.9	19.4	20.7	20.6	17.1	13.8	9.9	6.0	(22)	
	5%	DB	3.2	4.1	10.1	18.8	25.0	27.4	29.3	28.6	23.3	15.8	9.6	5.3	(23)	
		MCWB	2.3	2.8	6.0	11.6	16.8	19.0	20.3	19.6	16.0	12.4	8.4	4.6	(24)	
	10%	DB	1.8	2.7	7.8	16.6	22.9	25.5	27.3	26.4	20.8	13.6	8.1	3.5	(25)	
		MCWB	1.0	1.6	4.9	10.4	15.8	18.2	19.5	18.3	14.8	11.1	7.2	2.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.6	6.4	10.0	16.0	20.4	21.8	23.0	22.7	19.2	16.4	12.0	8.3	(27)	
		MCDB	6.5	8.1	15.1	23.5	26.8	28.1	29.7	30.4	26.3	19.7	13.2	8.9	(28)	
	2%	WB	3.6	4.3	8.0	13.7	18.9	20.6	22.0	21.4	17.7	14.4	10.2	6.1	(29)	
		MCDB	4.3	5.4	11.5	19.1	25.3	26.7	28.6	28.8	24.2	17.3	11.2	6.6	(30)	
	5%	WB	2.3	2.9	6.5	12.4	17.7	19.8	21.1	20.4	16.5	12.8	8.7	4.7	(31)	
		MCDB	3.1	4.1	9.5	17.3	23.4	25.9	27.1	27.0	21.8	15.4	9.5	5.3	(32)	
	10%	WB	1.1	1.8	5.1	11.2	16.6	18.9	20.3	19.4	15.4	11.3	7.2	2.7	(33)	
		MCDB	1.6	2.6	7.3	15.8	21.9	24.1	25.7	25.0	19.7	13.3	8.1	3.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.4	5.5	7.0	9.5	10.4	9.8	10.0	10.8	9.6	7.8	4.6	4.1	(35)	
		MCDBR	4.0	6.3	11.0	12.9	13.1	12.5	12.9	14.5	13.7	11.5	6.8	4.8	(36)	
	5% WB	MCWBR	3.7	4.8	6.6	6.3	5.7	5.2	4.8	5.7	6.1	6.8	5.2	4.2	(37)	
		MCWBR	3.9	5.7	10.0	11.3	11.8	11.2	11.4	12.8	12.1	10.4	5.8	4.6	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.306	0.338	0.369	0.431	0.406	0.394	0.426	0.428	0.396	0.361	0.342	0.309	(40)	
		taud	2.170	2.102	2.167	2.124	2.276	2.331	2.258	2.236	2.286	2.350	2.270	2.209	(41)	
	Ebn at Noon	Ebn at Noon	678	757	814	803	848	861	824	799	782	738	635	609	(42)	
		Edh at Noon	73	104	119	140	126	120	128	124	106	82	68	61	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.82	1.65	3.00	4.28	5.61	5.94	5.73	4.96	3.39	1.85	0.73	0.54	(44)	
	RadStd	RadStd	0.14	0.28	0.45	0.43	0.42	0.50	0.41	0.35	0.43	0.32	0.11	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	+0.62	N/A	N/A	+1.05	+0.75	+0.63	N/A	-182	+97	N/A
(47)	Regional (8 neighbors)	+0.65	N/A	N/A	+0.99	+0.57	+0.56	-118	-171	+101	+46

Nomenclature: See separate page